Da Dome Design Feature Justification

Feature	What it is	Design Rationale	Supporting Evidence (Interview Quotes)	Participant	Related Theme
Dome-shaped physical structure	A freestanding, visually distinctive dome (cardboard in the current prototype), tall and wide enough to be seen and gathered around from any side, placed within the festival grounds.	Almost every participant's default strategy for reuniting with a separated group was to nominate a fixed, memorable physical object ('a landmark') to return to a tree, a bar, a pole, a tent because phones routinely fail. Da Dome is built to function as a large, unusual, easy to describe structure that a group can nominate as 'meet at the dome' without needing signal to do so.	"we'll say, okay, like, we'll meet at said landmark... a big a big tree or like a certain type of bar or a merch stand" (P1) "we were just sitting at the bench... it was under a tree actually... they would also... look for the tree" (P2) "we always come up with like a safe plan being like, if anyone's phone runs out of battery or the reception's really bad, just meet at this point" (P8) "I guess like having different stations, resting points... so we can find each other... more like checkpoints" (P2)	P1, P2, P3, P8	Landmark Coordination; Network and Phone Unreliability
Fully physical, non-phone-dependent interaction	The dome is a self-contained hardware object (ESP32, NeoPixels, sensors) - no app, login, or network connection is required to use or find it.	Phone and network failure was one of the most consistent complaints across the sample, participants described calls, texts, and location sharing failing 'all the time' in dense crowds, sometimes forcing improvised workarounds (walkie talkies, totems, human chains). Any coordination or connection feature that depends on a live connection inherits this failure mode. By being a physical object, Da Dome works identically whether or not anyone's phone has signal.	"Yeah, all the time it fails all the time... you don't go to a music festival expecting to have service" (P1) "Often there's no signal, actually... calling, texting just doesn't work" (P4) "I'm planning on getting walkie-talkies because the cell service... tend[s] to cut out, but if you have your own walkie-talkie, it runs on its own network" (P7) "At BTV, yes, because there was no phone service... with so many people the service is often pretty bad" (P5)	P1, P2, P4, P5, P6, P7, P8	Network and Phone Unreliability; Physical Coordination Tactics
Multiple instrument 'tracks' (drums, keys, guitar, bass)	The dome's surface is divided into several distinct tactile tracks, each mapped to a different instrument, so a full 'song' only comes together when different people contribute different parts at once.	Festival groups are not monolithic as participants described repeatedly splitting into shifting sub groups by stage or artist preference, then reconverging. A single instrument or single user object would only mirror the solo/pair pattern, a multi-track instrument mirrors the actual social structure observed, a loose cluster of people, each doing their own thing, whose individual contributions only become a shared 'song' when they're near each other.	"if there are different artists playing at the same time, the group splits up and whoever prefers each artist goes their separate way, then maybe meets up for a different one" (P5) "it splits off into like 2 or 3 little groups... you kind of all just go off and do your own thing" (P6) "I don't mind going off by myself... at the very least, you usually do find your mates" (P1)	P1, P5, P6, P7	Loose Group Coupling
One hand is enough to activate a track	Only one participant needs to place a hand on a track for that instrument to sound, there is no minimum group size and no requirement to speak to anyone to take part.	Comfort with stranger contact varied sharply across the sample. Some participants love approaching people but hate being approached, others called themselves 'shy' and worried about 'overstepping boundaries', while a couple were simply not interested in meeting anyone new at a festival. Da Dome allows anyone to connect with eachother whetehr it is a stranger or friendd to cater for this	"I actually hate when people approach me... but I love approaching people" (P5) "I have such like a shy personality sometimes... I don't want to overstep boundaries or make people uncomfortable" (P8) "I'm okay with it, but I don't like it. I don't like other people talking to me at music festivals" (P4) "No, I tend not to chat with strangers... I'm not really interested in making friends at a concert" (P3)	P3, P4, P5, P8	Personal Space & Approach Comfort
Feedback scales with number of hands (brighter lights, louder mix, alternate notes)	Each additional hand on a track makes the lights on that track brighter and the instrument louder in the mix, and can trigger alternate notes so the dome visibly and audibly 'grows' as more people join in	Participants repeatedly described the pull of festivals in terms of collective atmosphere rather than one to one conversation, feeling 'part of a bigger thing' when a crowd sings or moves together, and singling out performers who could 'conduct' a crowd as the most memorable moments. Scaling the dome's output with participation recreates that effect in miniature and in the visitor's own hands, it turns joining in into a visible, non-verbal invitation to 'come put your hand here too' rather than something that has to be asked for out loud	"the most I could probably do with, like, stranger interaction is probably what Jacob Collier does with the conducting of the choir of the crowd... you feel like you're a part of... a bigger thing... you work together to be part of a song" (P3) "everyone is just so happy... within 3 seconds we'd made a friend and we were dancing under their umbrella" (P6) "a good performance is what makes things memorable" — torches, staging, crowd energy (P4)	P1, P3, P4, P6	Shared atmosphere
Playful, game-like framing (dome you physically dance/interact around)	The dome is approached and played rather than 'used', it is closer to a piece of playground equipment or a sensory installation than a utility device.	When asked directly, participants responded positively to the idea of a structured, playful tool for meeting people, describing it as something that would let them 'come out of my comfort zone', and independently requesting 'more games' as a way to interact between sets. One participant also described a near-identical existing installation (a walk-in sensory 'dust dome' with music and blindfolded dancing) as one of her favourite moments of a festival, which directly validates a dome shaped, playful, physical installation as a format festival-goers already enjoy and respond well to.	"something that's almost like an icebreaker would definitely make me come out of my comfort zone" (P8) "Would you be open to, like, more games set up for you to interact with other people... That sounds cool" (P7) "I took my shoes off and went into like the dust dome... everyone had the eye masks on... to feel the music and dance... it was cool because you got to like laugh about it" (P6)	P6, P7, P8	Interactive Installation Precedent; Stranger interaction
Short, drop-in/drop-out interaction (no fixed session length)	There's no start/end sequence to complete meaning a person can put a hand on the dome for a few seconds or a few minutes and leave at any point with no penalty or unfinished task.	Participants were explicit that any between-set activity has to fit inside a genuinely short window gaps between artists average 20-30 minutes, most of which is already spent on toilet/food breaks or walking between stages and that they wouldn't use anything that ate meaningfully into that time. A drop-in, drop-out interaction (rather than, say, a multi-step game) respects that constraint directly.	"definitely not long. I don't think long activities would be possible because sometimes the break between certain artists that are only like on average 20 to 30 minutes... by the time you get to the activity... a lot of that time's already taken away" (P8) "when there ever there's downtime... I always appreciate sitting down, having a drink and eating food to just rest my legs" (P8)	P8	Downtime / Between-Set Activity